

Data Science and Deep Learning — Practical Notes

A compact technical reference document designed for demonstrating document question answering with a Retrieval-Augmented Generation (RAG) system.

1. Introduction to Data Science

Data Science combines statistics, programming, machine learning, and domain knowledge to extract useful information from data. A typical data science workflow includes problem definition, data collection, data cleaning, exploratory data analysis, feature engineering, model development, evaluation, and deployment. The quality of the final model depends heavily on the quality and relevance of the data used during these stages.

2. Machine Learning

Machine learning allows a computer system to learn patterns from data and make predictions or decisions without explicitly programming every rule. Supervised learning uses labeled examples, while unsupervised learning works with data that does not have target labels. Common supervised algorithms include linear regression, logistic regression, decision trees, random forests, support vector machines, and neural networks. Clustering and dimensionality reduction are common unsupervised learning techniques.

3. Artificial Neural Networks

An artificial neural network is a computational model inspired by interconnected neurons. A basic neural network contains an input layer, one or more hidden layers, and an output layer. Each connection has a weight, and neurons commonly apply an activation function to a weighted combination of their inputs. During training, the network adjusts its weights to reduce a loss function.

4. Activation Functions

Activation functions introduce non-linearity into neural networks. ReLU, or Rectified Linear Unit, outputs zero for negative inputs and the input itself for positive inputs. Sigmoid maps values into a range between zero and one and is often used for binary probability outputs. Softmax converts a vector of scores into a probability distribution and is commonly used for multi-class classification. The choice of activation function affects how information moves through the network.

5. Backpropagation and Optimization

Backpropagation calculates how much each trainable parameter contributed to the prediction error. The calculated gradients are then used by an optimization algorithm to update model parameters. Gradient descent is a fundamental optimization method. Popular variants include Adam and RMSprop. Learning rate controls the size of parameter updates and is an important hyperparameter during training.

6. Convolutional Neural Networks

Convolutional Neural Networks, or CNNs, are especially useful for image data. A convolutional layer applies learnable filters to local regions of an image and can detect patterns such as edges, textures,

and shapes. Pooling layers can reduce spatial dimensions and computation. Deeper CNN layers can learn increasingly complex visual representations. CNNs are widely used for image classification, object detection, segmentation, and other computer vision tasks.

7. Recurrent Neural Networks

Recurrent Neural Networks, or RNNs, are designed for sequential information. An RNN maintains a hidden state that carries information from previous time steps. This makes RNNs useful for tasks involving text, time series, and other ordered data. Standard RNNs can have difficulty learning long-term dependencies because of vanishing or exploding gradients.

8. LSTM and GRU

Long Short-Term Memory networks, or LSTMs, extend RNNs with gates that regulate the flow of information. The input, forget, and output gates help an LSTM retain or discard information over time. Gated Recurrent Units, or GRUs, use a simpler gating mechanism with fewer parameters. Both LSTMs and GRUs are commonly used for sequence modeling and can handle longer dependencies better than a basic RNN.

9. Autoencoders

An autoencoder is a neural network that learns to represent input data in a compressed latent representation and then reconstruct the original input. It normally consists of an encoder and a decoder. Autoencoders can be used for dimensionality reduction, feature learning, anomaly detection, and image denoising. In image denoising, a noisy image can be provided as input while the corresponding clean image is used as the training target.

10. Model Evaluation

Classification models can be evaluated using accuracy, precision, recall, F1-score, and a confusion matrix. Accuracy measures the proportion of correct predictions. Precision measures how many predicted positive cases are actually positive, while recall measures how many actual positive cases are identified. F1-score combines precision and recall using their harmonic mean. The appropriate metric depends on the application and the cost of different types of errors.

11. Overfitting and Regularization

Overfitting occurs when a model learns training data too closely and performs poorly on unseen data. Common techniques for reducing overfitting include collecting more training data, data augmentation, dropout, early stopping, and weight regularization. Validation data can be used to monitor generalization during model development. A good model should learn useful patterns rather than memorize individual training examples.

12. Natural Language Processing

Natural Language Processing, or NLP, focuses on computational processing of human language. Typical NLP tasks include text classification, sentiment analysis, named entity recognition, machine translation, summarization, and question answering. Text must generally be converted into numerical representations before it can be processed by machine learning models. Modern systems often use learned embeddings to represent semantic relationships between words, sentences, or documents.

13. Embeddings

An embedding is a numerical vector representation of information such as text. Semantically similar pieces of text tend to have embeddings that are close together in vector space. Sentence-transformer models can create dense embeddings for sentences or document chunks. Embeddings are useful for semantic search because a query can be compared with stored document vectors to identify relevant information even when the wording is not identical.

14. Retrieval-Augmented Generation

Retrieval-Augmented Generation, or RAG, combines information retrieval with language generation. A document is first loaded and divided into smaller chunks. Each chunk is converted into an embedding and stored in a vector index. When a user asks a question, the question is embedded and compared with document embeddings. The most relevant chunks are retrieved and supplied as context to a language model, which generates an answer based on that context. RAG is useful for question answering over custom documents because the system can retrieve information from the source instead of relying only on a model's internal knowledge.

15. Vector Databases and Similarity Search

A vector store maintains vector representations and supports similarity search. FAISS is a library for efficient similarity search and clustering of dense vectors. Cosine similarity and inner product are commonly used to compare embeddings. A typical retrieval system selects the top-k chunks with the highest similarity to the query. The retrieved chunks can then be passed to a reranker or directly used as context for answer generation.

16. Practical RAG Workflow

A practical document question-answering pipeline can be organized into seven stages: document ingestion, text chunking, embedding creation, vector storage, query processing, context retrieval, and answer generation. Good chunking is important because chunks that are too small may lose context, while chunks that are too large may reduce retrieval precision. Retrieval quality should be checked with representative questions before evaluating the final generated answers.

17. Applications

Data science and deep learning techniques are used in healthcare, agriculture, finance, retail, manufacturing, education, cybersecurity, and many other areas. RAG systems can support enterprise knowledge assistants, technical documentation search, research assistants, customer support tools, and question answering over private documents. The choice of model, retrieval method, and evaluation metric should be based on the requirements of the application.

18. Key Takeaways

Data science involves a complete workflow from data preparation to model deployment. Deep learning models learn representations through multiple layers of trainable parameters. CNNs are effective for spatial data, while RNNs, LSTMs, and GRUs are useful for sequential data. Autoencoders learn compact representations and can reconstruct or denoise inputs. RAG combines semantic retrieval with language generation and is particularly useful when answers need to be grounded in custom documents.

Sample Questions for Document QA

1. What is the purpose of an activation function in a neural network?
2. How does a CNN process image data?
3. What problem do LSTMs address in standard RNNs?
4. What is an autoencoder and where can it be used?
5. What is the difference between precision and recall?
6. What is overfitting and how can it be reduced?
7. What are embeddings used for in semantic search?
8. What are the main stages of a RAG pipeline?
9. Why is text chunking important in a RAG system?
10. How does FAISS help a document question-answering system?